



Operations Research

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Exercise Sheet 2

Demo Exercises

Exercise D2.1. *Storage*

A storage room is to be equipped with shelves. The room has a total of $72m^2$ of floor space available for shelves. There are two types of shelves to choose from. The first shelf costs 10€, requires $6m^2$ of floor space, and provides $8m^3$ of storage capacity. The second shelf costs 20€, requires $8m^2$ of floor space, and provides $12m^3$ of storage capacity. How many shelves of each type would you buy to maximize the storage capacity if your budget is 140€? Determine the optimal solution using the Simplex algorithm.

Exercise D2.2. *Simplex-Algorithmus*

Consider the following Linear Program (LP):

$$\begin{array}{ll}\text{Max} & 2x_1 - x_2 \\ \text{s.t.} & -x_1 + x_2 \geq 2 \\ & x_1 + x_2 \leq 4 \\ & x \in \mathbb{R}_{\geq 0}^2\end{array}$$

- Solve the LP graphically.
- Transform the LP into standard form.
- Is the LP in canonical form? Justify.
- Compute an optimal basic solution for the LP using the Simplex algorithm, utilizing the Big-M method.
- Mark the feasible basic solutions found in part (d) on your graphical solution from part (a).

Exercise D2.3. *Simplex-Algorithmus*

Solve the following linear optimization problem using the Simplex algorithm:

$$\begin{array}{ll}\text{Max} & \\ \text{s.t.} & 4x_1 + x_2 = z \\ & x_1 + x_2 \leq 1 \\ & x_2 \leq 1 \\ & x_1, x_2 \in \mathbb{R}\end{array}$$